

Formalizing the Law in x.

Anonymous authors

Tutorial v1



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 - ▶ Example: Help people find out if they are eligible for social benefits



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 - ▶ Example: Help people find out if they are eligible for social benefits

Set-up: **lawyers and computer scientists collaborate** to read/write a formalization of the law.

Today's session

Our goal: Understand how well you (lawyers & computer scientists) can understand Lex code.

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1. Lex Tutorial + Q&A (40')
2. **Break! (5')**
3. Pre-task survey, Tasks 1+2 (35')
4. **Break! (5')**
5. Task 3, Post-task survey (33')

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5. Task 3, Post-task survey (33')

You should have: A tablet; a cheatsheet; a copy of these slides; a copy of the example program; a folded sheet of paper; scrap paper; a pen.

Tutorial Part I

Logic background

A model of legally relevant actions over time

Consider a webshop that has to ensure the privacy of its users.

Imagine writing down all legally relevant facts as they happen over time:

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2025-03-20 08:01:00	Alice creates an account on www.webshop.ch

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2025-03-20 08:01:00	Alice creates an account on www.webshop.ch
2025-03-20 08:02:00	Bob connects to www.webshop.ch

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2025-03-20 08:01:00	Alice creates an account on www.webshop.ch
2025-03-20 08:02:00	Bob connects to www.webshop.ch
2025-03-20 08:02:05	Bob clicks 'accept all cookies' on www.webshop.ch
2025-03-20 08:04:00	Bob creates an account on www.webshop.ch

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2025-03-20 08:01:00	Alice creates an account on www.webshop.ch
2025-03-20 08:02:00	Bob connects to www.webshop.ch
2025-03-20 08:02:05	Bob clicks 'accept all cookies' on www.webshop.ch
2025-03-20 08:04:00	Bob creates an account on www.webshop.ch
2025-03-20 09:00:00	Bob connects to www.webshop.ch (again) www.webshop.ch uses Bob's data to show him personalized ads

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2025-03-20 08:01:00	Alice creates an account on www.webshop.ch
2025-03-20 08:02:00	Bob connects to www.webshop.ch
2025-03-20 08:02:05	Bob clicks 'accept all cookies' on www.webshop.ch
2025-03-20 08:04:00	Bob creates an account on www.webshop.ch
2025-03-20 09:00:00	Bob connects to www.webshop.ch (again) www.webshop.ch uses Bob's data to show him personalized ads
2025-03-20 09:12:00	Alice connects to www.webshop.ch (again) www.webshop.ch uses Alice's data to show her personalized ads

A model of legally relevant actions over time

Consider a webshop that has to ensure the privacy of its users.

Imagine writing down all legally relevant facts as they happen over time:

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2025-03-20 08:00:01	<code>Connect</code> ("Alice", "www.webshop.ch")
2025-03-20 08:00:05	<code>RejectCookies</code> ("Alice", "www.webshop.ch")
2025-03-20 08:01:00	<code>CreateAccount</code> ("Alice", "www.webshop.ch")
2025-03-20 08:02:00	<code>Connect</code> ("Bob", "www.webshop.ch")
2025-03-20 08:02:05	<code>AcceptCookies</code> ("Bob", "www.webshop.ch")
2025-03-20 08:04:00	<code>CreateAccount</code> ("Bob", "www.webshop.ch")
2025-03-20 09:00:00	<code>Connect</code> ("Bob", "www.webshop.ch"), <code>UseData</code> ("www.webshop.ch", "Bob", "advertisement")
2025-03-20 09:12:00	<code>Connect</code> ("Alice", "www.webshop.ch"), <code>UseData</code> ("www.webshop.ch", "Alice", "advertisement")

A logical description of legal properties (1)

We may be interested to check all cases in which

“Some website uses a user’s data for advertisement as the user connects to the website.”

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E.g., this is the case with

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2025-03-20 09:12:00  Connect("Alice", "www.webshop.ch"),  
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How can we express this in a precise, mathematical way?

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How can we express this in a precise, mathematical way?

Solution: `Connect` (user, website) **AND** `UseData` (website, user, "advertisement")

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How can we express this in a precise, mathematical way?

Solution: `Connect` (user, website) **AND** `UseData` (website, user, "advertisement")

In this formula, user and website are **variables** that can match any user or website.

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More examples:

“Some user connects to a website or accepts the cookies.”

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“Bob rejects the cookies on some website or Bob does not accept the cookies on that website.”

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More examples:

“Some user connects to a website or accepts the cookies.”

→

`Connect` (user, website) OR `AcceptCookies` (user, website)

“Bob rejects the cookies on some website or Bob does not accept the cookies on that website.”

→

`RejectCookies` ("Bob", website) OR NOT `AcceptCookies` ("Bob", website)

A logical description of legal properties over time (1)

We may be interested to check all cases in which

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2025-03-20 09:12:00  Connect("Alice", "www.webshop.ch"),
                      UseData("www.webshop.ch", "Alice", "advertisement")
```

But if we write

`UseData (website, user, "advertisement") AND RejectCookies (user, website)`

we cannot detect the above because the rejection happens *before* data is used, not at the same time!

A logical description of legal properties over time (2)

Solution: have a way to explicitly say

“Some website uses a user’s data for advertisement but the user rejected the cookies in the past or present.”

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More examples:

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More examples:

“Some website uses a user’s data for advertisement but the user will reject the cookies in the future.”

→

`UseData (website, user, "advertisement") AND EVENTUALLY RejectCookies (user, website)`

A logical description of legal properties over time (2)

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We write

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More examples:

“Some website uses a user’s data for advertisement but the user will reject the cookies in the future.”

→

`UseData` (website, user, "advertisement") **AND** **EVENTUALLY** `RejectCookies` (user, website)

“Some website uses a user’s data for advertisement but the user did never accept the cookies.”

A logical description of legal properties over time (2)

Solution: have a way to explicitly say

“Some website uses a user’s data for advertisement but the user rejected the cookies in the past or present.”

We write

`UseData` (website, user, "advertisement") **AND** **ONCE** `RejectCookies` (user, website)

More examples:

“Some website uses a user’s data for advertisement but the user will reject the cookies in the future.”

→

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“Some website uses a user’s data for advertisement but the user did never accept the cookies.”

→

`UseData` (website, user, "advertisement") **AND** **NOT ONCE** `AcceptCookies` (user, website)

Tutorial Part II

Legal background

What does a law contain?

A law:

- ▶ Talks about things that exist in the world (individuals, companies, buildings, user data...).
- In generally, a single law only talks about *specific* things (privacy law: humans computers...).

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The Swiss flag shows a Swiss cross on a square background.

— *Coat of Arms Protection Act, Article 3(1)*

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- ▶ Introduces *legal definitions*, e.g.



The Swiss flag shows a Swiss cross on a square background.

— *Coat of Arms Protection Act, Article 3(1)*

- ▶ Define obligations of persons



Any person who uses a public sign must prove that he or she is allowed do so.

— *Coat of Arms Protection Act, Article 19*

An important distinction

The law contains rules.

Some rules **create obligations** ('obligation rules') while others **define facts** ('constitutive rules').

Examples: how the Swiss flag can be used vs. what it looks like.

The structure of the law

Very frequent structure: general rule followed by a number of exceptions.



- 1. Processing of personal data revealing racial or ethnic origin, political opinions, [...] shall be prohibited.*
- 2. Paragraph 1 shall not apply if one of the following applies:*
 - (a) the data subject has given explicit consent...;*
 - ...*
 - (j) ...*

— General Data Protection Regulation, Article 9

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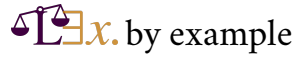


1. *Processing of personal data revealing racial or ethnic origin, political opinions, [...] shall be prohibited.*
2. *Paragraph 1 shall not apply if one of the following applies:*
 - (a) *the data subject has given explicit consent...;*
 - ...
 - (j) ...

— *General Data Protection Regulation, Article 9*

An exception can be defined anywhere, potentially several hundred pages away!

Tutorial Part III



Excerpts from the General Data Protection Regulation (GDPR)



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(a) the data subject has given consent to the processing of his or her personal data for one or more specific purposes;

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1. Where point (a) of Article 6(1) applies, [...] the processing of the personal data of a child shall be lawful where the child is at least 16 years old.

— Art. 8 GDPR

First step: Types

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Art. 5(1): Personal **data** shall be:

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Art. 6(1): Processing shall be lawful only if and to the extent that at least one of the following applies:

(a) the **data subject** has given consent to the processing of his or her personal data for one or more specific **purposes**;

Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the personal **data** of a **child** [**data subject**] shall be lawful where the **child** is at least 16 years old.

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+ Implicitly, the data is processed by some **entity** (= company) as part of some processing **activity**

In Lex, we define each kind of persons/objects as a *type*:

```
type activity
type data
type data_subject
type purpose
type entity
```

Second step: Events and predicates (1)

What does the law say about these persons/objects?

Art. 5(1): **Personal data** shall be:

(a) processed lawfully, fairly and in a transparent manner in relation to the **data subject**;

Art. 6(1): Processing shall be **lawful** only if and to the extent that at least one of the following applies:

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Art. 6(1): Processing shall be **lawful** only if and to the extent that at least one of the following applies:

(a) the **data subject** has given consent to the processing of his or her personal data for one or more specific **purposes**;

Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the **personal data** of a **child** [**data subject**] shall be **lawful** where the **child** is at least 16 years old.

The law discusses some events (= actions, verbs) as well as predicates (= relations, adjectives).

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The law discusses some events (= actions, verbs) as well as predicates (= relations, adjectives).

Example: In Lex, we declare the predicate for 'personal data' as follows:

```
predicate PersonalData
  "" "Data {d} is personal data of data subject {ds}" ""
  d   : data
  ds  : data_subject
```

Second step: Events and predicates (2)

Art. 5(1): **Personal data** shall be:

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Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the **personal data** of a **child [data subject]** shall be **lawful** where the **child** is at least 16 years old.

predicate IsLawful

```
"""Activity {a} is lawful"""
```

```
a : activity
```

event Process

```
"""Entity {p} processes data {d}  
as part of activity {a}"""
```

```
p : entity
```

```
a : activity
```

```
d : data
```


Second step: Events and predicates (2)

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predicate IsLawful

"""Activity {a} is lawful"""

a : activity

event Process

"""Entity {p} processes data {d}
as part of activity {a}"""

p : entity

a : activity

d : data

event GiveConsent

"""Data subject {ds} consents
that their data is used by
{c} for purpose {p}"""

ds : data_subject

p : purpose

c : entity

predicate IsMinor

"""Data subject {ds} is < 16"""

ds : data_subject

Third step: Rules (1)

Art. 5(1): **Personal data** shall be:

(a) **processed lawfully, fairly** and **in a transparent manner** in relation to the **data subject**;

Third step: Rules (1)

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(a) **processed lawfully, fairly** and **in a transparent manner** in relation to the **data subject**;

This is an **obligation rule**.

Third step: Rules (1)

Art. 5(1): **Personal data** shall be:

(a) **processed lawfully, fairly** and **in a transparent manner** in relation to the **data subject**;

This is an **obligation rule**. In Lex:

```
article "5"  
paragraph "1"  
point "a"  
  
rule  
  whenever  
    Process(p, a, d)  
    PersonalData(d, ds)  
  oblige  
    IsLawful(a)  
    IsFair(a)  
    IsTransparent(a)
```

Third step: Rules (2)

Art. 6(1): Processing shall be **lawful** only if and to the extent that at least one of the following applies:

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(a) the **data subject** has given consent to the processing of his or her personal data for one or more specific **purposes**;

This is a **constitutive rule**.

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(a) the **data subject** has given consent to the processing of his or her personal data for one or more specific **purposes**;

This is a **constitutive rule**. In Lex:

```
article "6"  
paragraph "1"  
point "a"  
  
rule  
  whenever  
    Process(c, a, d)  
    PersonalData(d, ds)  
    ONCE GiveConsent(ds, p, c)    # GiveConsent can be *before* Process  
  constitute  
    IsLawful(a)
```

Third step: Rules (3)

Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the **personal data** of a **child** [data subject] shall be **lawful** where the **child** is at least 16 years old.

Third step: Rules (3)

Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the **personal data** of a **child** [data subject] shall be **lawful** where the **child** is at least 16 years old.

This is an **exception rule**.

Third step: Rules (3)

Art. 8(1): Where point (a) of Article 6(1) applies, [...] the processing of the **personal data** of a **child** [data subject] shall be **lawful** where the **child** is at least 16 years old.

This is an **exception rule**. In Lex:

```
article "8"  
paragraph "1"  
  
rule  
  whenever  
    Process(c, a, d)  
    PersonalData(d, ds)  
    IsMinor(ds)  
  except  
    article "6" paragraph "1" point "a"
```

Third step: Additional syntax for rules

“Whenever data is processed, consent must have been given within a week (before the processing).”

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“Whenever data is processed, consent must have been given within a week (before the processing).”

```
rule
  whenever
    Process(c, a, d)
    PersonalData(d, ds)
  oblige once within 7d
    GiveConsent(ds, p, c)
```

Third step: Additional syntax for rules

“Whenever data is processed, consent must have been given within a week (before the processing).”

```
rule
  whenever
    Process(c, a, d)
    PersonalData(d, ds)
  oblige once within 7d
    GiveConsent(ds, p, c)
```

“Whenever data is processed, consent must be given within five days (after the processing).”

Third step: Additional syntax for rules

“Whenever data is processed, consent must have been given within a week (before the processing).”

```
rule
  whenever
    Process(c, a, d)
    PersonalData(d, ds)
  oblige once within 7d
    GiveConsent(ds, p, c)
```

“Whenever data is processed, consent must be given within five days (after the processing).”

```
rule
  whenever
    Process(c, a, d)
    PersonalData(d, ds)
  oblige eventually within 5d
    GiveConsent(ds, p, c)
```

Documentation

⇒ **Open HTML documentation file**

Thank you for your attention!

Q&A

Break (5')

Break (5')

Q1: Demographics (5')

Q1: Demographics (5')

Q2: Main task (10')

Q2: Main task (10')

Q3: SUS survey (5')

Q3: SUS survey (5')

Q4: Reading tasks (15')

Q4: Reading tasks (15')

Break (5')

Break (5')

Q5.A–C: Auditing task – individual phase (12’)

Q5.D–G: Auditing task – pair phase (12’)

Q6.A–D: Post-task survey (5')